Homework Assignments 3

Chapter 6 Simple Linear Regression

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Contents

報告概述 / Overview

這次作業出自 Montgomery 教科書第 6 章 *Simple Linear Regression*，共兩題：Exercise 6-5 與 6-11。資料是某混凝土研究裡 15 筆觀測，自變數是 compressive strength (x)，反應變數是 intrinsic permeability (y)，模型為

$$y = \beta_0 + \beta_1 x + \varepsilon.$$

- **Exercise 6-5 (a)-(k)**：估計係數、殘差、SS_E 與變異數、係數標準誤、$SS_T = SS_R + SS_E$ 分解、R^2、係數 t 檢定、ANOVA F 檢定、95% CI、模型適合度診斷、樣本相關係數檢定。
- **Exercise 6-11 (a)-(d)**：在 $x_0 = 2.1$ 求平均反應、99% CI、99% PI，並比較兩個區間的寬度。

依助教規定，過程數值算到第三位、最終答案四捨五入到第三位。套件載入與共用設定（`fmt3`、`print_header`）統一放在腳本最前段；本報告每小題直接呈現該題的統計計算程式碼與 console 結果，繪圖的 `ggplot` 樣板碼為精簡版面而省略，只保留最終圖檔。

Exercise 6-5

> **6-5.** An article in *Concrete Research* ("Near Surface Characteristics of Concrete: Intrinsic Permeability," Vol. 41, 1989) presented data on compressive strength x and intrinsic permeability y of various concrete mixes and cures. 15 observations are given.
>
> (a) Estimate the intercept β_0 and slope β_1. Write the estimated regression line.
>
> (b) Compute the residuals.
>
> (c) Compute SS_E and estimate the variance.
>
> (d) Find the standard error of the estimated slope and intercept coefficients.
>
> (e) Show that $SS_T = SS_R + SS_E$.
>
> (f) Compute the coefficient of determination R^2. Comment on the value.
>
> (g) Use a t-test for significance of the intercept and slope at $\alpha = 0.05$. Give the P-values and comment.
>
> (h) Construct the ANOVA table and test for significance of regression using the P-value. Relate to part (g).
>
> (i) Construct 95% CIs on the intercept and slope. Relate to parts (g) and (h).
>
> (j) Perform model adequacy checks. Does the model fit adequately?
>
> (k) Compute the sample correlation coefficient and test for its significance at $\alpha = 0.05$. Relate to (g) and (h).

資料輸入與模型配適

```r
# --- 2. Input the actual dataset from Exercise 6-5 ---
# 資料照題目原樣 hard-code，15 筆，順序不可改
strength <- c(3.1, 4.5, 3.4, 2.5, 2.2, 1.2, 5.3, 4.8,
              2.4, 3.5, 1.3, 3.0, 3.3, 3.2, 1.8)
permeability <- c(33.0, 31.0, 34.9, 35.6, 36.1, 39.0, 30.1, 31.2,
                  35.7, 31.9, 37.3, 33.8, 32.8, 31.6, 37.7)

# response 是 permeability，explanatory 是 strength
(Concrete <- data.frame(
  strength = strength,
  permeability = permeability
))
print_subhead("Summary of the Concrete data")
summary(Concrete)
#>    strength permeability
#> 1       3.1         33.0
#> 2       4.5         31.0
#> 3       3.4         34.9
#> 4       2.5         35.6
#> 5       2.2         36.1
#> 6       1.2         39.0
#> 7       5.3         30.1
#> 8       4.8         31.2
#> 9       2.4         35.7
#> 10      3.5         31.9
#> 11      1.3         37.3
#> 12      3.0         33.8
#> 13      3.3         32.8
#> 14      3.2         31.6
#> 15      1.8         37.7
#>
#> --- Summary of the Concrete data ---
#>     strength      permeability
```

```
#>  Min.   :1.200   Min.   :30.10
#>  1st Qu.:2.300   1st Qu.:31.75
#>  Median :3.100   Median :33.80
#>  Mean   :3.033   Mean   :34.11
#>  3rd Qu.:3.450   3rd Qu.:35.90
#>  Max.   :5.300   Max.   :39.00
```

```
# --- 3. Fit the simple linear regression model ---
# 主模型：permeability ~ strength
(lm.Concrete <- lm(permeability ~ strength, Concrete))
model_summary <- summary(lm.Concrete)
print(model_summary)
# olsrr 的整合報表，老師範例會印
ols_regress(permeability ~ strength, Concrete)
#>
#> Call:
#> lm(formula = permeability ~ strength, data = Concrete)
#>
#> Coefficients:
#> (Intercept)     strength
#>      40.554       -2.123
#>
#>
#> Call:
#> lm(formula = permeability ~ strength, data = Concrete)
#>
#> Residuals:
#>     Min      1Q  Median      3Q     Max
#> -2.1595 -0.6203  0.2173  0.8184  1.5652
#>
#> Coefficients:
#>             Estimate Std. Error t value Pr(>|t|)
#> (Intercept)  40.5536     0.7509  54.004  < 2e-16 ***
#> strength     -2.1232     0.2313  -9.181 4.81e-07 ***
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#>
#> Residual standard error: 1.038 on 13 degrees of freedom
#> Multiple R-squared:  0.8664, Adjusted R-squared:  0.8561
#> F-statistic: 84.28 on 1 and 13 DF,  p-value: 4.805e-07
#>
#>                         Model Summary
#> --------------------------------------------------------------
#> R                       0.931       RMSE                0.966
#> R-Squared               0.866       MSE                 0.933
#> Adj. R-Squared          0.856       Coef. Var           3.042
#> Pred R-Squared          0.828       AIC                47.532
#> MAE                     0.797       SBC                49.656
#> --------------------------------------------------------------
#>  RMSE: Root Mean Square Error
#>  MSE: Mean Square Error
#>  MAE: Mean Absolute Error
#>  AIC: Akaike Information Criteria
#>  SBC: Schwarz Bayesian Criteria
#>
#>                                ANOVA
#> -------------------------------------------------------------------
#>               Sum of
#>              Squares        DF    Mean Square      F          Sig.
```

```
#> -----------------------------------------------------------------
#> Regression      90.759          1          90.759     84.285     0.0000
#> Residual        13.999         13           1.077
#> Total          104.757         14
#> -----------------------------------------------------------------
#>
#>                                   Parameter Estimates
#> ------------------------------------------------------------------------------------------
#>         model        Beta    Std. Error      Std. Beta         t        Sig        lower        upper
#> ------------------------------------------------------------------------------------------
#> (Intercept)     40.554         0.751                       54.004     0.000     38.931     42.176
#>    strength     -2.123         0.231         -0.931       -9.181     0.000     -2.623     -1.624
#> ------------------------------------------------------------------------------------------
```

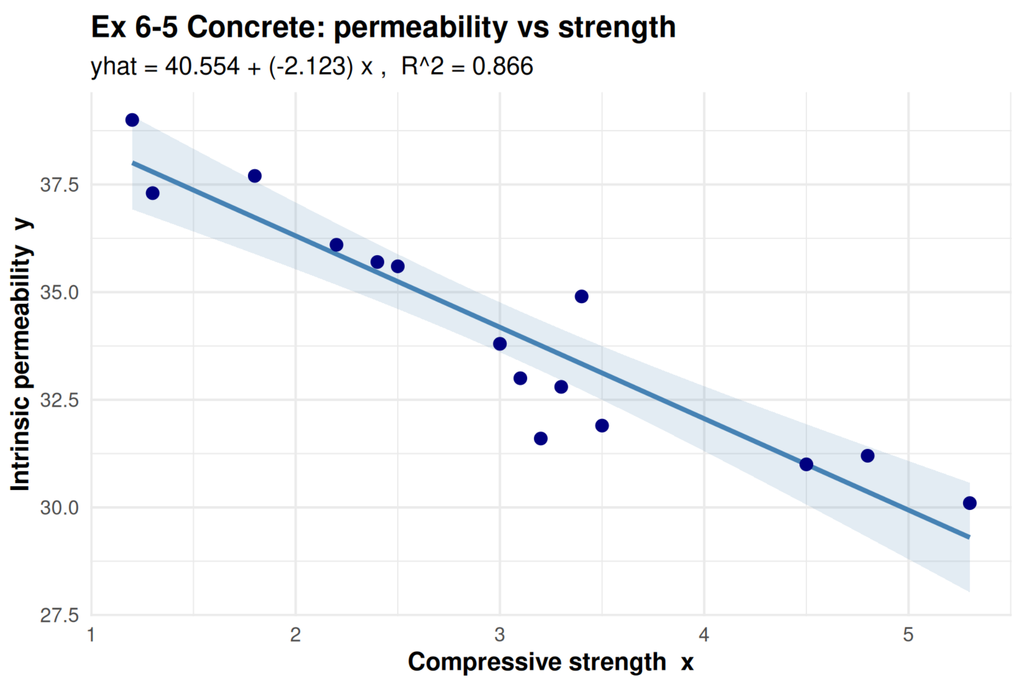


Figure 1: Ex 6-5 散布圖與最小平方回歸線（含 95% 平均反應信賴帶）。X 軸為 compressive strength x、Y 軸為 intrinsic permeability y；負斜率對應 $\hat{\beta}_1 = -2.123$。

(a)-(d) 估計、殘差、SS_E、變異數、標準誤

```
# --- 4. Exercise 6-5(a)-(d): estimates, residuals, SSE, variance, SEs ---
print_header("EXERCISE 6-5 (a)-(d)")

# (a) 截距、斜率與回歸線
b <- coef(lm.Concrete)
cat("(a) Estimated regression line\n")
cat(sprintf("beta0_hat = %.3f , beta1_hat = %.3f\n", b[1], b[2]))
cat(sprintf("yhat = %.3f + (%.3f) * x\n", b[1], b[2]))

# (b) 殘差表，題目要求看得到每一筆，不能只給圖
(Residual_df <- data.frame(
  Observation  = 1:nrow(Concrete),
  strength     = Concrete$strength,
  permeability = Concrete$permeability,
  fitted       = round(fitted(lm.Concrete), 3),
  residual     = round(resid(lm.Concrete), 3)
))
```

```
# (c) SSE 與變異數估計 (MSE)
SSE <- sum(resid(lm.Concrete)^2)
MSE <- SSE / lm.Concrete$df.residual          # variance 的估計值 sigma^2_hat
cat(sprintf("\n(c) SSE = %.3f\n", SSE))
cat(sprintf("    df (residual) = %d\n", lm.Concrete$df.residual))
cat(sprintf("    sigma^2_hat (MSE) = %.3f\n", MSE))
cat(sprintf("    residual standard error sigma_hat = %.3f\n", sqrt(MSE)))

# (d) 兩個估計係數的 standard error
se_coef <- model_summary$coefficients[, "Std. Error"]
cat(sprintf("\n(d) SE(beta0_hat) = %.3f\n", se_coef[1]))
cat(sprintf("    SE(beta1_hat) = %.3f\n", se_coef[2]))
#>
#> ============================================================
#>  EXERCISE 6-5 (a)-(d)
#> ============================================================
#> (a) Estimated regression line
#> beta0_hat = 40.554 , beta1_hat = -2.123
#> yhat = 40.554 + (-2.123) * x
#>    Observation strength permeability fitted residual
#> 1            1      3.1          33.0 33.972   -0.972
#> 2            2      4.5          31.0 30.999    0.001
#> 3            3      3.4          34.9 33.335    1.565
#> 4            4      2.5          35.6 35.246    0.354
#> 5            5      2.2          36.1 35.883    0.217
#> 6            6      1.2          39.0 38.006    0.994
#> 7            7      5.3          30.1 29.301    0.799
#> 8            8      4.8          31.2 30.362    0.838
#> 9            9      2.4          35.7 35.458    0.242
#> 10          10      3.5          31.9 33.123   -1.223
#> 11          11      1.3          37.3 37.794   -0.494
#> 12          12      3.0          33.8 34.184   -0.384
#> 13          13      3.3          32.8 33.547   -0.747
#> 14          14      3.2          31.6 33.759   -2.159
#> 15          15      1.8          37.7 36.732    0.968
#>
#> (c) SSE = 13.999
#>     df (residual) = 13
#>     sigma^2_hat (MSE) = 1.077
#>     residual standard error sigma_hat = 1.038
#>
#> (d) SE(beta0_hat) = 0.751
#>     SE(beta1_hat) = 0.231
```

(e)-(h) 平方和分解、R^2、t 檢定、ANOVA

```
# --- 5. Exercise 6-5(e)-(h): ANOVA, R-squared, t-tests, F-test ---
print_header("EXERCISE 6-5 (e)-(h)")

# (e) 證明 SST = SSR + SSE
ybar <- mean(Concrete$permeability)
SST <- sum((Concrete$permeability - ybar)^2)
SSR <- sum((fitted(lm.Concrete) - ybar)^2)
cat("(e) Sum-of-squares decomposition\n")
cat(sprintf("SST = %.3f\n", SST))
cat(sprintf("SSR = %.3f\n", SSR))
cat(sprintf("SSE = %.3f\n", SSE))
cat(sprintf("SSR + SSE = %.3f  (應該等於 SST = %.3f)\n", SSR + SSE, SST))
```

```r
# (f) 判定係數 R^2
R2 <- model_summary$r.squared
cat(sprintf("\n(f) R^2 = %.3f\n", R2))
cat(sprintf("    亦即 strength 解釋了 permeability 約 %.1f%% 的變異。\n", 100 * R2))

# (g) 截距與斜率的 t 檢定 (alpha = 0.05)
print_subhead("(g) t-tests for intercept and slope, alpha = 0.05")
print(model_summary$coefficients)
t_tab <- model_summary$coefficients
cat(sprintf("intercept: t = %.3f, p = %.3g\n", t_tab[1, 3], t_tab[1, 4]))
cat(sprintf("slope    : t = %.3f, p = %.3g\n", t_tab[2, 3], t_tab[2, 4]))
cat(" 兩個 p-value 都 < 0.05，截距與斜率都顯著。\n")

# (h) ANOVA table + regression F 檢定
print_subhead("(h) ANOVA table")
(aov_tab <- anova(lm.Concrete))
Fstat <- aov_tab[1, "F value"]
Fp    <- aov_tab[1, "Pr(>F)"]
cat(sprintf("F = %.3f, p = %.3g\n", Fstat, Fp))
# 對照：simple linear regression 裡 F 應該等於 slope t 的平方
cat(sprintf(" 檢查 F = t^2 ? F = %.3f, t_slope^2 = %.3f\n",
            Fstat, t_tab[2, 3]^2))
#>
#> ============================================================
#>  EXERCISE 6-5 (e)-(h)
#> ============================================================
#> (e) Sum-of-squares decomposition
#> SST = 104.757
#> SSR = 90.759
#> SSE = 13.999
#> SSR + SSE = 104.757  (應該等於 SST = 104.757)
#>
#> (f) R^2 = 0.866
#>     亦即 strength 解釋了 permeability 約 86.6% 的變異。
#>
#> --- (g) t-tests for intercept and slope, alpha = 0.05 ---
#>              Estimate Std. Error   t value     Pr(>|t|)
#> (Intercept) 40.553642  0.7509317 54.004433 1.106512e-16
#> strength    -2.123179  0.2312658 -9.180685 4.805472e-07
#> intercept: t = 54.004, p = 1.11e-16
#> slope    : t = -9.181, p = 4.81e-07
#> 兩個 p-value 都 < 0.05，截距與斜率都顯著。
#>
#> --- (h) ANOVA table ---
#> Analysis of Variance Table
#>
#> Response: permeability
#>           Df Sum Sq Mean Sq F value    Pr(>F)
#> strength   1 90.759  90.759  84.285 4.805e-07 ***
#> Residuals 13 13.999   1.077
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#> F = 84.285, p = 4.81e-07
#> 檢查 F = t^2 ? F = 84.285, t_slope^2 = 84.285
```

(i) 係數的 95% 信賴區間

```
# --- 6. Exercise 6-5(i): coefficient confidence intervals ---
print_header("EXERCISE 6-5 (i): 95% CIs for intercept and slope")
(ci_coef <- confint(lm.Concrete, level = 0.95))
cat(sprintf("95%% CI for beta0: [%.3f , %.3f]\n", ci_coef[1, 1], ci_coef[1, 2]))
cat(sprintf("95%% CI for beta1: [%.3f , %.3f]\n", ci_coef[2, 1], ci_coef[2, 2]))
cat(" 兩個 CI 都不含 0，跟 (g)(h) 的顯著結論一致。\n")
#>
#> ============================================================
#>  EXERCISE 6-5 (i): 95% CIs for intercept and slope
#> ============================================================
#>                 2.5 %    97.5 %
#> (Intercept) 38.931353 42.175932
#> strength    -2.622798 -1.623559
#> 95% CI for beta0: [38.931 , 42.176]
#> 95% CI for beta1: [-2.623 , -1.624]
#> 兩個 CI 都不含 0，跟 (g)(h) 的顯著結論一致。
```

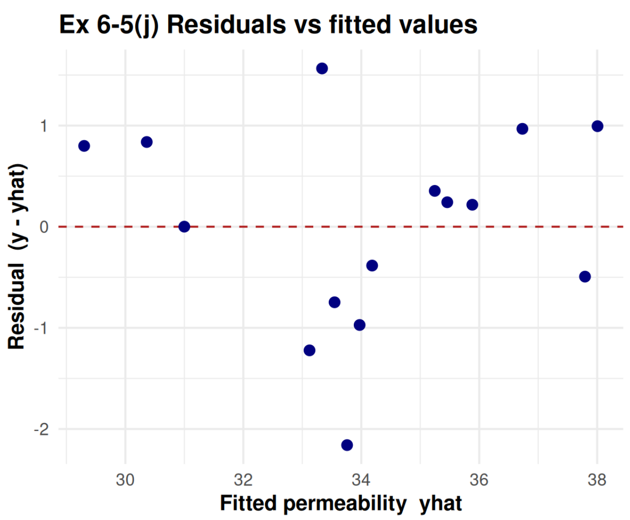
(j) 模型適合度診斷

```
# --- 7. Exercise 6-5(j): model adequacy checks ---
print_header("EXERCISE 6-5 (j): model adequacy checks")

# 殘差的常態性、等變異、離群、影響點一起看
print_subhead("Breusch-Pagan test (constant variance)")
print(ols_test_breusch_pagan(lm.Concrete))
print_subhead("Normality tests on residuals")
print(ols_test_normality(lm.Concrete))
print_subhead("Outlier test (studentized residuals, Bonferroni)")
print(outlierTest(lm.Concrete))
print_subhead("Influence measures summary")
print(summary(influence.measures(lm.Concrete)))

# 標準 lm 四連圖存成 png
png("figures/6-5j_lm_diagnostics.png", width = 1500, height = 1200, res = 170)
opar <- par(mfrow = c(2, 2), oma = c(0, 0.3, 1.3, 0))
plot(lm.Concrete)
par(opar)
invisible(dev.off())
#>
#> ============================================================
#>  EXERCISE 6-5 (j): model adequacy checks
#> ============================================================
#>
#> --- Breusch-Pagan test (constant variance) ---
#>
#>  Breusch Pagan Test for Heteroskedasticity
#>  -----------------------------------------
#>  Ho: the variance is constant
#>  Ha: the variance is not constant
#>
#>                  Data
#>  ----------------------------------------
#>  Response : permeability
#>  Variables: fitted values of permeability
#>
#>         Test Summary
#>  ------------------------------
```

```
#>  DF              =    1
#>  Chi2            =    0.08570349
#>  Prob > Chi2     =    0.769712
#>
#> --- Normality tests on residuals ---
#> -----------------------------------------------
#>         Test             Statistic       pvalue
#> -----------------------------------------------
#> Shapiro-Wilk               0.969          0.8434
#> Kolmogorov-Smirnov         0.1213         0.9609
#> Cramer-von Mises           0.8167         0.0058
#> Anderson-Darling           0.2189         0.8012
#> -----------------------------------------------
#>
#> --- Outlier test (studentized residuals, Bonferroni) ---
#> No Studentized residuals with Bonferroni p < 0.05
#> Largest |rstudent|:
#>     rstudent unadjusted p-value Bonferroni p
#> 14 -2.583729           0.023933      0.35899
#>
#> --- Influence measures summary ---
#> Potentially influential observations of
#>   lm(formula = permeability ~ strength, data = Concrete) :
#>
#>    dfb.1_ dfb.strn dffit cov.r   cook.d hat
#> 7  -0.43   0.57     0.64  1.51_*  0.21   0.32
#> 14 -0.15  -0.10    -0.70  0.52_*  0.17   0.07
#>        dfb.1_     dfb.strn      dffit      cov.r     cook.d        hat
#> 7  -0.4290918  0.57074230  0.6409732 1.5055670 0.2075685 0.32185430
#> 14 -0.1536935 -0.09941244 -0.6981557 0.5199249 0.1696454 0.06804636
```



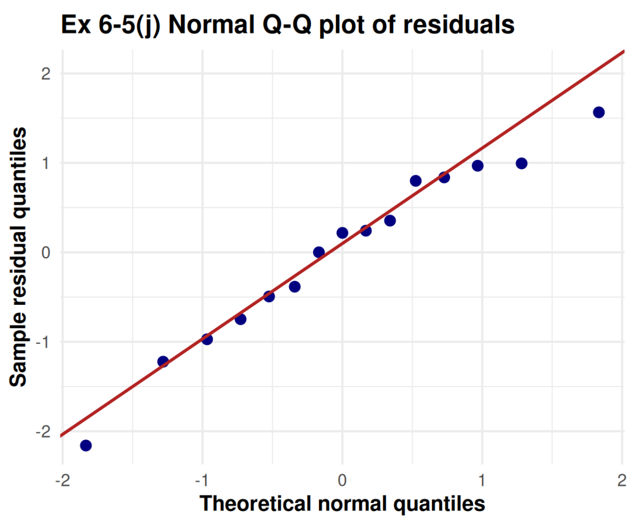


Figure 2: Ex 6-5(j) 殘差診斷。左：殘差對 fitted（檢查線性與等變異），點圍繞 0 無明顯曲度或喇叭口。右：殘差 Normal Q-Q（檢查常態），點貼合參考線。

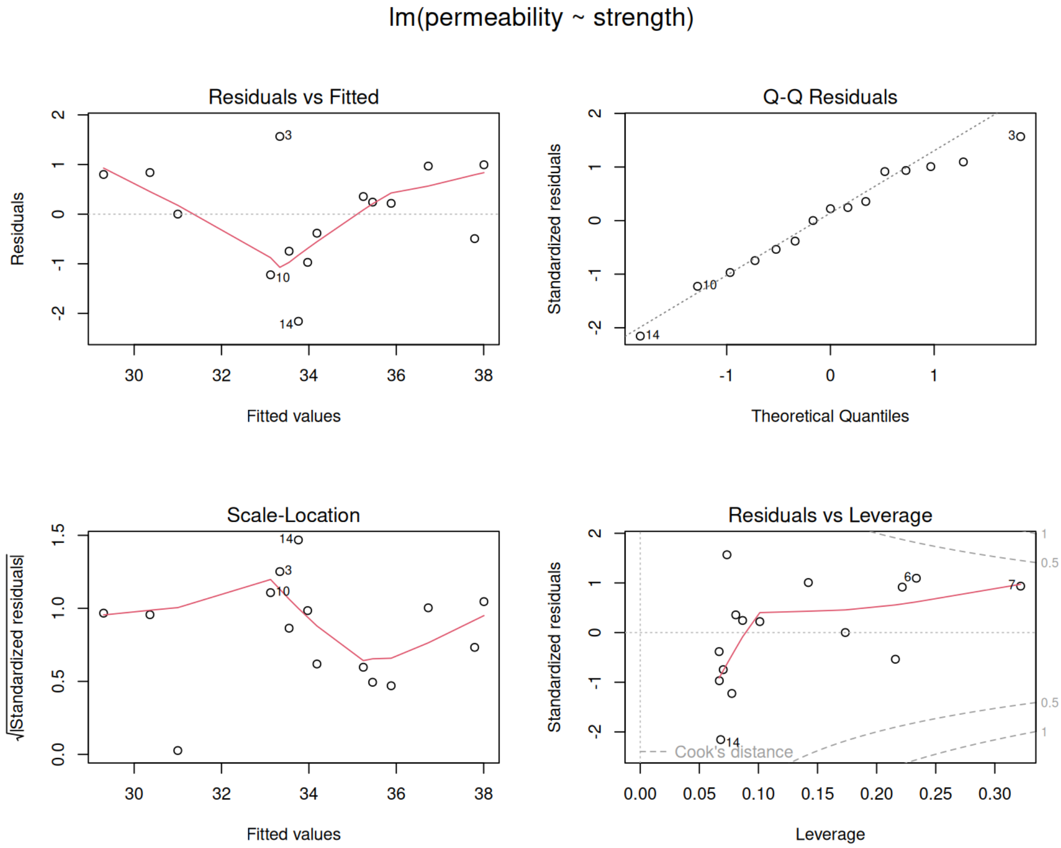


Figure 3: Ex 6-5(j) 標準 lm 四連圖：Residuals vs Fitted、Normal Q-Q、Scale-Location、Residuals vs Leverage。

(k) 樣本相關係數與檢定

```
# --- 8. Exercise 6-5(k): correlation coefficient and test ---
print_header("EXERCISE 6-5 (k): sample correlation and test")
(r_xy <- cor(Concrete$strength, Concrete$permeability))
ct <- cor.test(Concrete$strength, Concrete$permeability)
print(ct)
cat(sprintf("r = %.3f\n", r_xy))
cat(sprintf("correlation t = %.3f, p = %.3g\n", ct$statistic, ct$p.value))
# 三個檢定在 simple linear regression 應該同一個結論、同一個 p
cat(sprintf(" 檢查 r^2 = R^2 ? r^2 = %.3f, R^2 = %.3f\n", r_xy^2, R2))
#>
#> ============================================================
#>  EXERCISE 6-5 (k): sample correlation and test
#> ============================================================
#> [1] -0.930791
#>
#>  Pearson's product-moment correlation
#>
#> data:  Concrete$strength and Concrete$permeability
#> t = -9.1807, df = 13, p-value = 4.805e-07
#> alternative hypothesis: true correlation is not equal to 0
#> 95 percent confidence interval:
#>  -0.9771428 -0.7999540
#> sample estimates:
#>        cor
#> -0.930791
#>
#> r = -0.931
#> correlation t = -9.181, p = 4.81e-07
```

```
#> 檢查 r^2 = R^2 ? r^2 = 0.866, R^2 = 0.866
```

最終答案 Final Answer

- **(a)** $\hat{\beta}_0 = 40.554$、$\hat{\beta}_1 = -2.123$；回歸線 $\hat{y} = 40.554 - 2.123\,x$。
- **(b)** 殘差見上方殘差表（每筆 $e_i = y_i - \hat{y}_i$），最大正殘差在 Obs 3（+1.565）、最大負殘差在 Obs 14（−2.159）。
- **(c)** $SS_E = 13.999$（$df = 13$）；$\hat{\sigma}^2 = MSE = 1.077$，殘差標準誤 $\hat{\sigma} = 1.038$。
- **(d)** $SE(\hat{\beta}_0) = 0.751$、$SE(\hat{\beta}_1) = 0.231$。
- **(e)** $SS_T = 104.757 = SS_R + SS_E = 90.759 + 13.999$，分解成立。
- **(f)** $R^2 = 0.866$，strength 解釋了 permeability 約 86.6% 的變異，配適度高。
- **(g)** 截距 $t = 54.004$（$P \approx 1.11 \times 10^{-16}$）、斜率 $t = -9.181$（$P \approx 4.805 \times 10^{-7}$），都 < 0.05，兩係數皆顯著。
- **(h)** ANOVA $F = 84.285$、$P \approx 4.805 \times 10^{-7} < 0.05$，回歸顯著；且 $F = 84.285 = (-9.181)^2 = t^2_{\text{slope}}$，與 (g) 等價。
- **(i)** 95% CI：$\beta_0 \in [38.931, 42.176]$、$\beta_1 \in [-2.623, -1.624]$，都不含 0，與 (g)(h) 一致。
- **(j)** 線性、常態（Shapiro-Wilk $p = 0.843$）、等變異（Breusch-Pagan $p = 0.770$）都通過，無顯著離群（Obs 14 Bonferroni $p = 0.359$）、無高 Cook's distance。因 $n = 15$ 偏小不過度宣稱，整體模型配適足夠。
- **(k)** $r = -0.931$，$t = -9.181$、$P \approx 4.805 \times 10^{-7} < 0.05$，顯著負相關；$r^2 = 0.866 = R^2$，且與 (g)(h) 的 slope/regression 檢定結論、P-value 完全一致。

Exercise 6-11

> **6-11.** Consider the data and simple linear regression model in Exercise 6-5.
> (a) Find the mean permeability given that the strength is 2.1.
> (b) Compute a 99% CI on this mean response.
> (c) Compute a 99% PI on a future observation when the strength is equal to 2.1.
> (d) What do you notice about the relative size of these two intervals? Which is wider and why?

```
# --- 9. Exercise 6-11: mean response, 99% CI, 99% PI at x0 = 2.1 ---
print_header("EXERCISE 6-11: prediction at strength x0 = 2.1")

(newdata <- data.frame(strength = 2.1))

# 注意：predict 用具名參數比較不會踩到位置參數的雷（df 跟 level 容易搞錯）
# (a) 點估計 mean response
pred_fit <- predict(lm.Concrete, newdata)
cat(sprintf("(a) mean permeability at x0 = 2.1 : %.3f\n", pred_fit))

# (b) 平均反應的 99% CI
pred_ci <- predict(lm.Concrete, newdata, interval = "confidence", level = 0.99)
cat(sprintf("(b) 99%% CI  : [%.3f , %.3f]\n", pred_ci[2], pred_ci[3]))

# (c) 單一未來觀測的 99% PI
pred_pi <- predict(lm.Concrete, newdata, interval = "prediction", level = 0.99)
cat(sprintf("(c) 99%% PI  : [%.3f , %.3f]\n", pred_pi[2], pred_pi[3]))

# (d) 比較寬度
w_ci <- pred_ci[3] - pred_ci[2]
w_pi <- pred_pi[3] - pred_pi[2]
cat(sprintf("(d) CI width = %.3f , PI width = %.3f\n", w_ci, w_pi))
cat("PI 比 CI 寬，因為 PI 除了平均反應的估計誤差，還多加了單一未來觀測的隨機誤差。\n")


# ============================================================================
# Figures (ggplot)：軸標都給單位、標題粗體，方便列印閱讀
# ============================================================================
#>
#> ============================================================
#>  EXERCISE 6-11: prediction at strength x0 = 2.1
#> ============================================================
#>   strength
#> 1      2.1
#> (a) mean permeability at x0 = 2.1 : 36.095
#> (b) 99% CI  : [35.059 , 37.131]
#> (c) 99% PI  : [32.802 , 39.388]
#> (d) CI width = 2.073 , PI width = 6.586
#> PI 比 CI 寬，因為 PI 除了平均反應的估計誤差，還多加了單一未來觀測的隨機誤差。
```

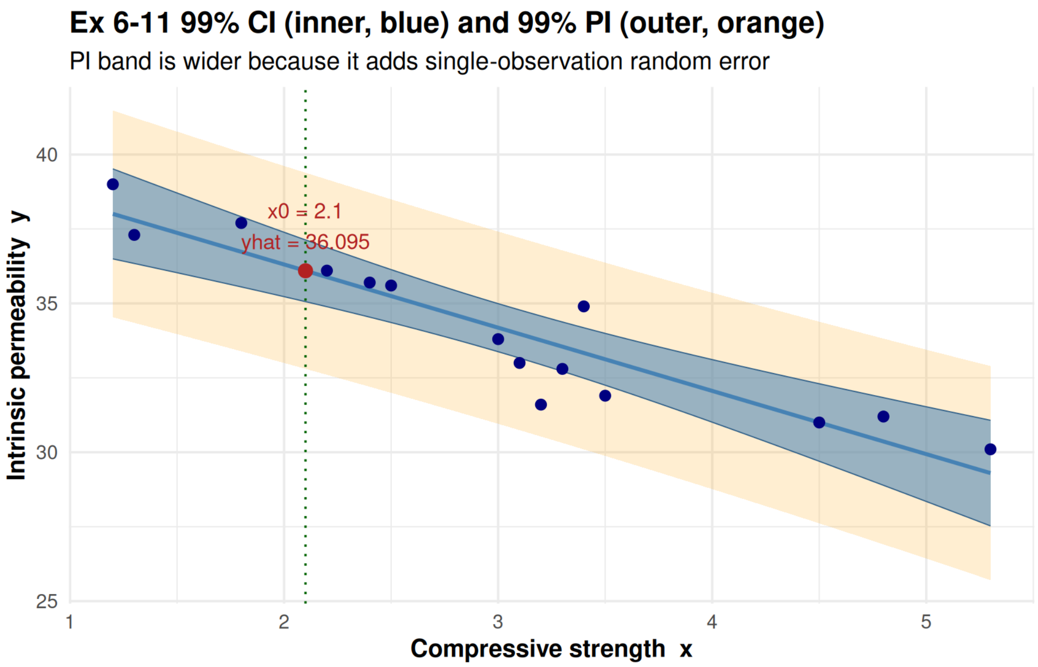


Figure 4: Ex 6-11 沿著 strength 的 99% CI（內層藍帶）與 99% PI（外層橘帶）。綠點線標出 $x_0 = 2.1$，紅點為點估計 $\hat{y} = 36.095$。PI 帶明顯比 CI 帶寬。

最終答案 Final Answer

- **(a)** 在 $x_0 = 2.1$，平均反應 $\hat{\mu}_{Y|x_0} = 40.554 - 2.123 \times 2.1 = 36.095$。
- **(b)** 99% CI（mean response）：$[35.059,\ 37.131]$，寬度 2.073。
- **(c)** 99% PI（future observation）：$[32.802,\ 39.388]$，寬度 6.586。
- **(d)** PI（寬 6.586）比 CI（寬 2.073）寬。因為 CI 只含平均反應的估計不確定性，PI 還要再加上單一未來觀測本身的隨機誤差 ε（多了 $\hat{\sigma}^2$ 一項），所以區間更寬。

結語 / Closing Notes

幾個這次有特別留意的點：

- 資料原樣保留：15 筆 strength / permeability 照題目順序 hard-code，沒有改動。
- 三個檢定一致：simple linear regression 裡 slope t 檢定、回歸 ANOVA F 檢定、相關係數檢定結論相同；數值上驗證了 $F = t^2 = 84.285$、$r^2 = R^2 = 0.866$、三者 P-value 都是 4.805×10^{-7}。
- 平方和分解：$SS_T = SS_R + SS_E$ 用數值核對過，只有四捨五入層級差異。
- **CI 與 PI** 的差別：6-11 的 PI 比 CI 寬，是因為 PI 多算了單一未來觀測的隨機誤差。
- 診斷不過度宣稱：$n = 15$ 樣本小，常態與等變異檢定大多通過（少數如 Cramer-von Mises 例外），但搭配 Q-Q 圖與其餘檢定，仍給出「足夠配適」這個保守結論。
- 精度：全篇最終答案四捨五入到第三位（助教規定）。

R 4.5.2 跑完所有計算，繪圖用 `ggplot2`（散布/殘差/Q-Q/CI-PI 帶）與 base R `plot(lm)` 四連圖；模型與檢定用 `lm`、`summary`、`anova`、`confint`、`predict`、`cor.test`、`olsrr`、`car`。